

IDEATHON ON VIKSHIT BHARAT:

Ideas, Innovation and Impact

Edited by
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**COMPARATIVE IN-SILICO MAPPING OF TP53 MUTATION
HOTSPOTS IN ELEPHANTS: A RESPONSIBLE
BIOINFORMATICS INNOVATION CONTRIBUTING TO CANCER
RESEARCH**

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❖ **ABSTRACT**

T*his study investigates the conservation of major human TP53 mutation hotspots in elephant canonical and retrogene sequences using a reproducible in-silico comparative genomics framework. Elephants exhibit unusually low cancer incidence despite large body size and lifespan, a phenomenon known as Peto's paradox. To explore possible molecular explanations, clinically significant human TP53 mutation hotspots (R175, G245, R248, R249, R273 and R282) were mapped onto African and Asian elephant TP53 protein sequences using protein-level multiple sequence alignment. Sequence datasets were retrieved from publicly available databases including NCBI and UniProt and analysed using bioinformatics tools such as*

MAFFT, Jalview, MEGA and Python-based scripts implemented in Google Colab. Results indicate strong conservation of key DNA-binding residues in canonical elephant TP53 sequences, whereas retrogene copies exhibit position-specific variability suggesting divergent evolutionary trajectories among duplicated genes. A composite scoring framework was applied to prioritise mutation hotspots based on mutation frequency, conservation and retrogene variability. The findings highlight the evolutionary stability of canonical TP53 residues and demonstrate the usefulness of transparent computational workflows for comparative genomic research.

Keywords: TP53; Responsible genomics; Research governance; Bioinformatics reproducibility; Elephant retrogenes; Ethical innovation

1. INTRODUCTION

The tumor suppressor gene **TP53** encodes the p53 protein, a transcription factor that plays a central role in maintaining genomic stability. p53 regulates several critical cellular processes including cell cycle arrest, DNA repair, senescence, and apoptosis in response to DNA damage. Because of its essential role in tumor suppression, TP53 is one of the most frequently mutated genes in human cancers. A large proportion of these mutations occur within the DNA-binding domain of the protein, particularly at specific hotspot residues such as **R175, G245, R248, R249, R273, and R282**. These residues are structurally and functionally important, as some directly interact with DNA while others maintain the structural stability of the protein.

An interesting biological observation related to cancer risk across species is known as **Peto's paradox**, which states that large and long-lived animals do not necessarily exhibit higher cancer incidence than smaller animals, despite having many more cells and longer lifespans. Elephants represent a notable example of this paradox. Although elephants possess a significantly larger body size and lifespan compared to humans, studies have shown that they exhibit relatively low cancer mortality. Previous genomic studies have revealed that elephants contain multiple copies of the TP53 gene, including a canonical gene and several retrogene copies. These

retrogenes are believed to have arisen through reverse transcription events and may contribute to enhanced tumor suppression mechanisms.

While earlier studies have primarily focused on the expansion of TP53 gene copy number in elephants and its potential role in increased apoptotic responses, detailed residue-level comparative analysis of **human TP53 mutation hotspots across elephant canonical and retrogene sequences remains limited**. Understanding whether these critical mutation sites are conserved or altered across species may provide insights into evolutionary constraints on tumor suppressor genes and the potential functional significance of duplicated TP53 variants.

Therefore, the present study aims to perform a **comparative in-silico mapping of major human TP53 mutation hotspots onto canonical and retrogene TP53 protein sequences in elephants**. By applying protein-level multiple sequence alignment and computational analysis using publicly available genomic datasets, this work evaluates the conservation and variability of key residues associated with human cancer mutations. The study also emphasizes methodological transparency and reproducibility by using well-documented bioinformatics tools and openly accessible databases. Through this approach, the research contributes to responsible computational genomics while providing a structured framework for prioritizing mutation hotspots for future experimental investigation.

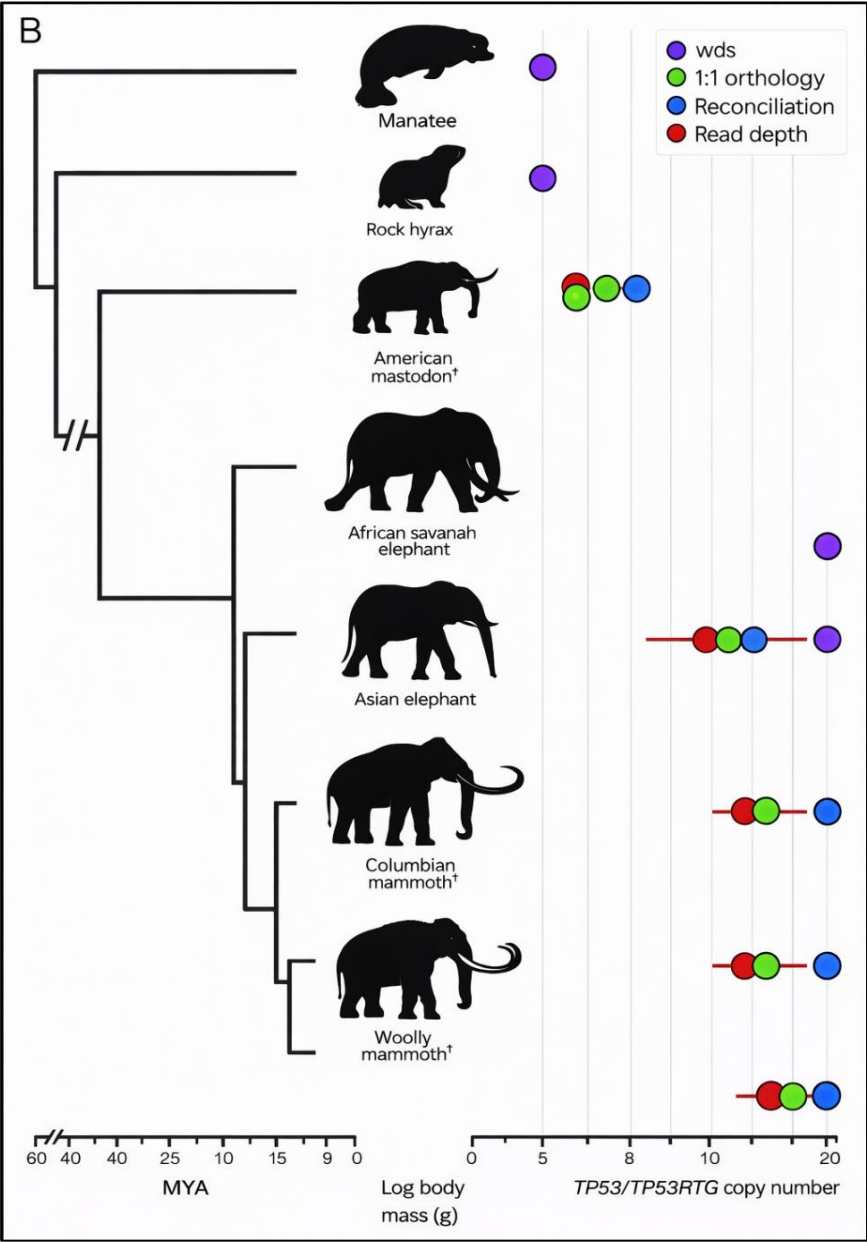


Figure 1. Comparative representation of TP53 copy number expansion in elephant species.

2. AIM AND OBJECTIVES

The primary aim of this study is to perform a structured in-silico comparative mapping of major human TP53 mutation hotspots across canonical and retrogene TP53 protein sequences in elephants. By analysing sequence-level conservation and divergence patterns, the study seeks to evaluate whether residues frequently mutated in human cancers are preserved in elephant TP53 variants or exhibit evolutionary variation.

To achieve this aim, the following objectives were defined:

1. To retrieve curated TP53 protein sequences of humans, African elephants (*Loxodonta africana*), and Asian elephants (*Elephas maximus*) from publicly accessible biological databases including NCBI and UniProt.
2. To perform protein-level multiple sequence alignment using established bioinformatics tools in order to ensure accurate positional correspondence of residues across species.
3. To identify and map major human TP53 mutation hotspots—including residues R175, G245, R248, R249, R273, and R282—within aligned elephant canonical and retrogene sequences.
4. To evaluate conservation and variability patterns of these hotspot residues among elephant TP53 variants, with particular attention to differences between canonical and retrogene sequences.
5. To implement a composite scoring framework that integrates mutation frequency, cross-species conservation, and retrogene variability in order to prioritise mutation hotspots for further biological investigation.

Through these objectives, the study aims to provide a reproducible computational framework for comparative genomic analysis while supporting responsible interpretation of sequence-based findings.

3. MATERIALS AND METHODS

3.1 Data Sources and Ethical Data Use

All sequence data used in this study were obtained from publicly accessible biological databases, primarily the National Center for Biotechnology Information (NCBI) and the UniProt Knowledgebase. Only curated reference protein sequences were selected to ensure

reliability and traceability of the dataset. No human patient data, clinical records, or personally identifiable genomic datasets were used in this analysis. All database access complied with publicly stated data usage policies, ensuring ethical and responsible use of genomic resources.

3.2 Dataset Construction

The dataset used for comparative analysis included the human TP53 canonical protein sequence, canonical TP53 sequences from African elephant (*Loxodonta africana*) and Asian elephant (*Elephas maximus*), and available elephant TP53 retrogene protein sequences reported in public databases. All sequences were downloaded in FASTA format and organised into a structured dataset for computational analysis. Duplicate entries and incomplete or truncated sequences were removed to maintain data quality and consistency.

3.3 Sequence Preprocessing and Quality Control

Prior to alignment, sequence preprocessing steps were performed to ensure data integrity. These included verification of sequence length consistency, removal of ambiguous amino acid characters, and confirmation that the DNA-binding domain region of TP53 was present in all selected sequences. When necessary, nucleotide sequences were translated into protein sequences using standard bioinformatics tools. These preprocessing steps ensured accurate residue-level comparison across species.

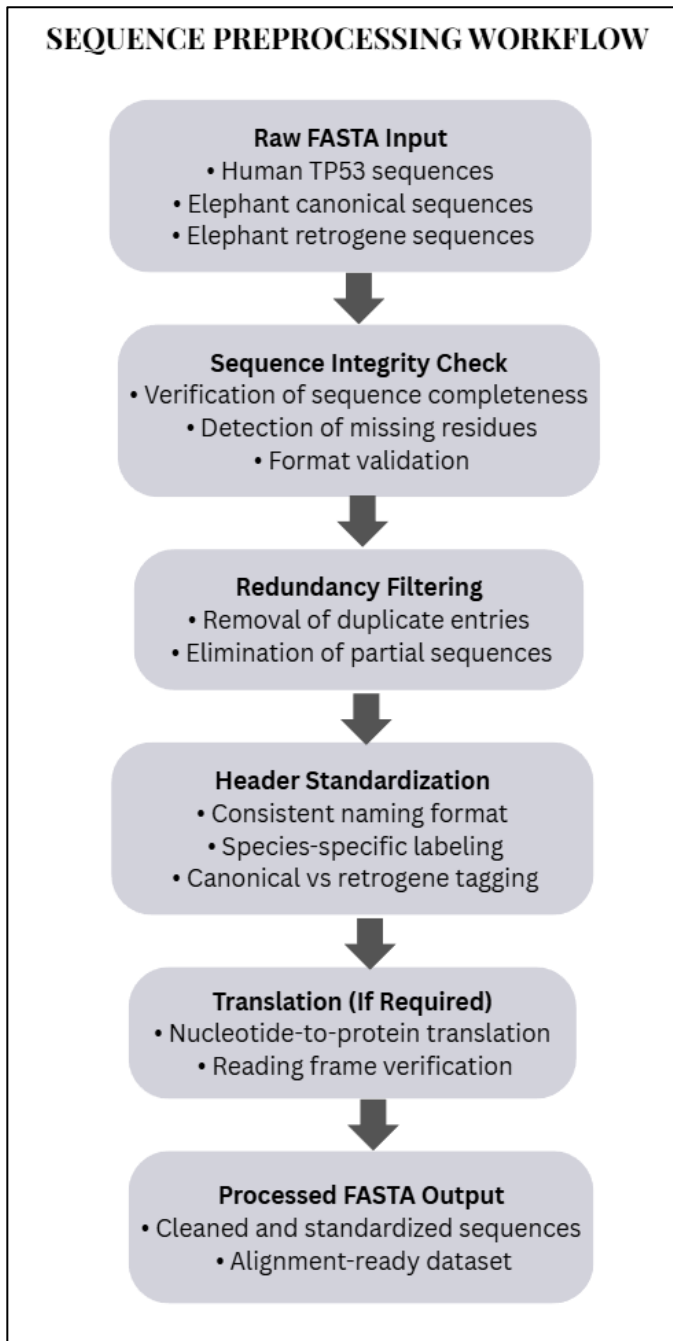


Figure 2. Sequence preprocessing and quality control workflow used for TP53 dataset preparation.

3.4 Multiple Sequence Alignment

Protein-level multiple sequence alignment (MSA) was performed using the MAFFT alignment algorithm to maintain structural correspondence within the TP53 DNA-binding domain. Protein-level alignment was preferred over nucleotide alignment to avoid codon frame inconsistencies and to allow direct comparison of amino acid residues. Alignment results were further inspected using Jalview to confirm correct positional correspondence of key residues and to evaluate conservation patterns across sequences.

3.5 Hotspot Residue Mapping

Six major human TP53 mutation hotspots frequently associated with cancer were selected for comparative analysis: R175, G245, R248, R249, R273, and R282. These hotspot residues were first identified in the human reference TP53 sequence and then mapped onto the aligned elephant canonical and retrogene sequences using the multiple sequence alignment output. Residues at each hotspot position were examined to determine whether they were conserved, conservatively substituted, or non-conservatively substituted.

3.6 Composite Scoring Framework

To systematically prioritise hotspot residues, a composite descriptive scoring approach was implemented. The scoring framework integrated three parameters: Human Mutation Frequency (HMF) derived from curated mutation databases, Cross-Species Conservation (CSC) representing residue identity between human and elephant canonical sequences, and Retrogene Variability Index (RVI) reflecting the proportion of retrogene sequences showing substitutions at a given position. The composite score was calculated using the formula:

$$\text{Composite Score} = (\text{HMF_norm} + \text{CSC_norm} + (1 - \text{RVI_norm})) / 3$$

This scoring method provides a structured approach for ranking mutation hotspots while remaining descriptive rather than predictive.

3.7 Computational Tools

All computational analyses were performed using widely accepted bioinformatics tools. Sequence alignment was conducted using MAFFT, alignment visualization and inspection were performed

using Jalview, and phylogenetic analysis was carried out using MEGA software. Additional data processing and scoring calculations were implemented using Python-based scripts in Google Colab, utilising libraries such as NumPy and Pandas. The use of openly available tools and clearly documented workflows ensures reproducibility of the analysis.

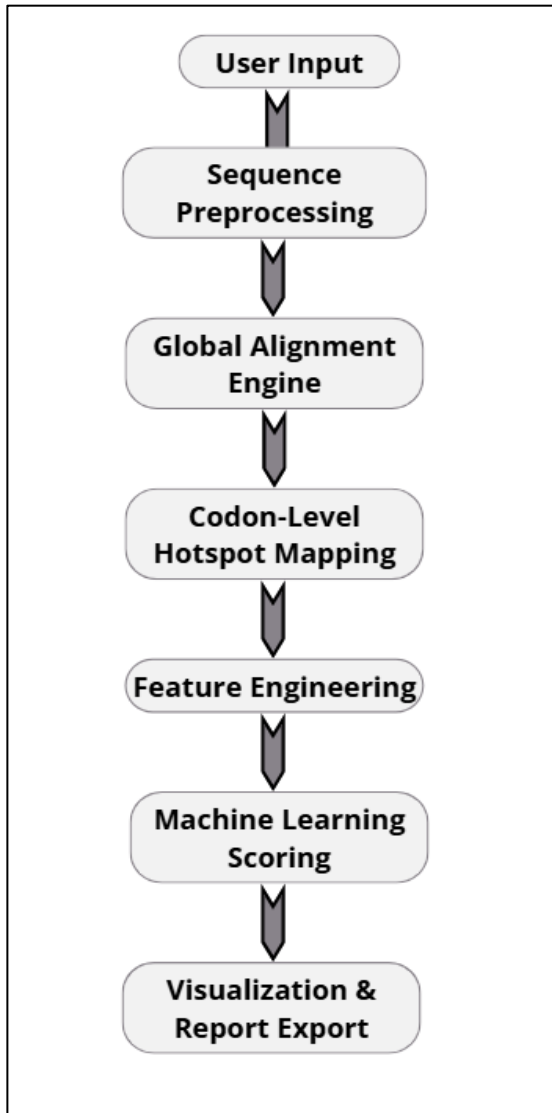


Figure 3. Overview of the EleProtect computational workflow for TP53 hotspot mapping.

4. RESULTS

4.1 Multiple Sequence Alignment

Protein-level multiple sequence alignment was performed using the human TP53 sequence as the positional reference. The alignment revealed high overall similarity between human and elephant canonical TP53 protein sequences, particularly within the DNA-binding domain, which spans approximately residues 100–300 of the protein. This region is known to contain several critical residues involved in DNA interaction and structural stability.

Inspection of the alignment showed minimal gap distribution in the DNA-binding domain, indicating strong structural conservation across species. The positions corresponding to the selected human TP53 mutation hotspots—R175, G245, R248, R249, R273, and R282—were accurately aligned in all canonical sequences, enabling reliable residue-level comparison between human and elephant proteins.

The alignment also confirmed that the structural region surrounding these hotspot residues remains largely conserved across species, supporting the functional importance of this domain.

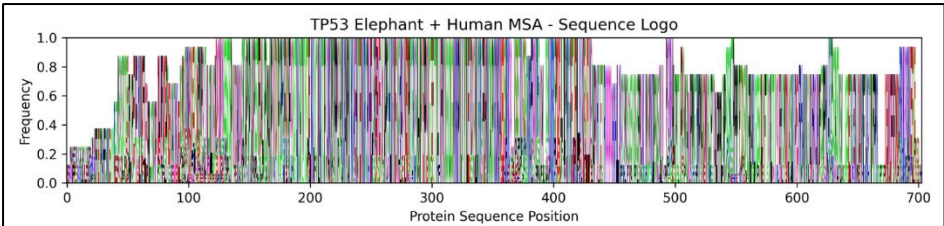


Figure 4. Multiple sequence alignment of human and elephant TP53 sequences highlighting conservation within the DNA-binding domain.

4.2 Conservation of Major TP53 Mutation Hotspots

Residue-level analysis of the selected mutation hotspots demonstrated strong conservation in canonical elephant TP53 sequences. Each of the six major human TP53 hotspot residues was found to be preserved in both African and Asian elephant canonical sequences.

Specifically, residue R175 remained conserved as arginine, while G245 remained glycine, and R248, R249, R273, and R282 were also conserved across the canonical sequences analysed. No non-conservative substitutions were observed at these positions in the canonical TP53 sequences of elephants.

This high level of conservation suggests that these residues are subject to strong evolutionary constraints, reflecting their structural and functional importance in the DNA-binding domain of the TP53 protein.

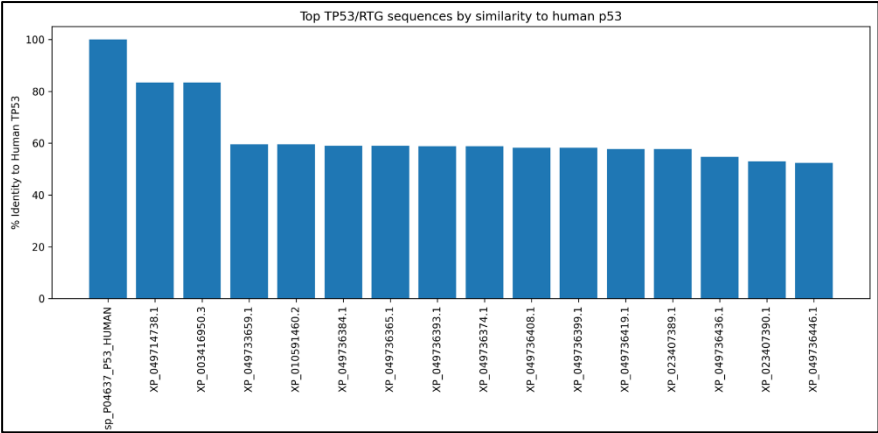


Figure 5. Residue-level mapping of major human TP53 mutation hotspots across canonical and retrogene elephant sequences.

4.3 Retrogene Variability Analysis

In contrast to canonical sequences, analysis of elephant TP53 retrogene variants revealed position-specific variability at several hotspot positions. While some retrogene copies retained identical residues to the canonical sequence, others displayed substitutions at selected hotspot sites.

The variability observed among retrogene sequences indicates that duplicated TP53 copies may have undergone divergent evolutionary processes, resulting in heterogeneous residue patterns across different retrogene variants. However, no universal disruptive mutation pattern was detected across all retrogene sequences,

suggesting that these copies may exhibit varying levels of functional retention or divergence.

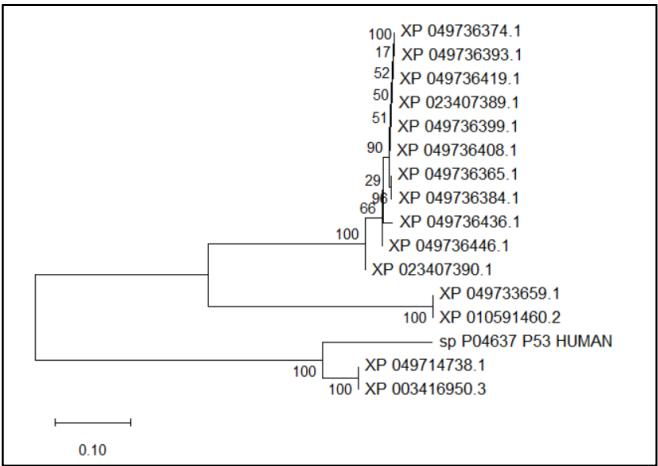


Figure 6. Phylogenetic relationships among human and elephant TP53 canonical and retrogene protein sequences.

4.4 Composite Hotspot Prioritisation

To prioritise mutation hotspots for further investigation, a composite scoring framework was applied that integrates human mutation frequency, cross-species conservation, and retrogene variability. This approach enables structured ranking of residues based on their potential biological relevance.

The analysis identified residues R175, G245, R248, and R273 as the highest-ranking hotspots. These positions combine high mutation frequency in human cancers with strong conservation in canonical elephant sequences and relatively limited variability among retrogene copies.

The composite scoring results provide a structured method for highlighting mutation sites that may warrant further investigation in future structural and functional studies.

Codon Position	Human (TP53)	African Elephant (Canonical)	Asian Elephant (Canonical)	Retrogene Copies	Conservation Value
175	R	R	R	R	Conserved
245	G	G	G	G	Conserved
248	R	R	R	R	Conserved
249	R	R	R	R	Conserved
273	R	R	R	R	Conserved
282	R	R	R	R	Conserved

Figure 7. Composite prioritisation of TP53 hotspot residues integrating mutation frequency, conservation, and retrogene variability.

5. DISCUSSION AND CONCLUSION

The present study provides a structured residue-level comparison of major human TP53 mutation hotspots across canonical and retrogene TP53 sequences in elephants. The results demonstrate strong conservation of the six well-established human mutation hotspots—R175, G245, R248, R249, R273, and R282—within canonical elephant TP53 sequences. These residues are located within the DNA-binding domain of the p53 protein and play an essential role in maintaining the structural stability and transcriptional activity of the tumor suppressor protein. Their preservation across species suggests that these residues are subject to strong evolutionary constraints and remain functionally important for maintaining genomic stability.

In contrast to canonical sequences, elephant TP53 retrogene variants exhibited position-specific variability at certain hotspot residues. This variability indicates that duplicated TP53 copies may have followed different evolutionary trajectories after their emergence through retrotransposition events. While some retrogene sequences retain residues identical to the canonical TP53 sequence, others show substitutions that may reflect relaxed selective pressure or partial functional divergence. However, sequence-level variation alone does not necessarily indicate functional gain or loss, and additional experimental studies would be required to determine the biological significance of these substitutions.

The implementation of the EleProtect composite scoring framework further enabled structured prioritisation of hotspot residues by integrating mutation frequency, cross-species conservation, and retrogene variability. This approach identified residues such as

R175, G245, R248, and R273 as high-priority positions for future investigation due to their combined evolutionary conservation and high mutation prevalence in human cancers. Importantly, the scoring framework is descriptive rather than predictive and is intended to guide further biological investigation rather than infer direct functional outcomes.

Overall, this study demonstrates how comparative in-silico genomic analysis can contribute to understanding evolutionary patterns in tumor suppressor genes while maintaining methodological transparency and responsible interpretation of computational findings. By combining publicly available genomic data with reproducible computational workflows, the research provides a foundation for future structural, transcriptomic, and experimental studies that may further clarify the role of TP53 variants in cancer resistance mechanisms across species.

6. FUTURE RESEARCH DIRECTIONS

Future research should extend the present sequence-based analysis by integrating additional computational and experimental approaches. Structural modeling and molecular dynamics simulations could provide deeper insights into how conserved and substituted residues influence the stability and function of the TP53 protein. Furthermore, transcriptomic studies may help determine whether elephant TP53 retrogenes are actively expressed and contribute to cellular tumor suppression mechanisms. Functional laboratory assays would also be required to evaluate the biological impact of residue-level variations observed in retrogene sequences. Expanding comparative genomic analysis to additional large and long-lived mammalian species may further improve our understanding of evolutionary mechanisms underlying cancer resistance.

7. ACKNOWLEDGEMENT

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of Bioinformatics for providing academic resources and computational support required for this study.

8 CONFLICT OF INTEREST

The authors declare that there is no conflict of interest regarding the publication of this research work.

9. DATA AVAILABILITY STATEMENT

All sequence datasets used in this study were obtained from publicly available biological databases, including the National Center for Biotechnology Information (NCBI) and the UniProt Knowledgebase. The datasets analyzed during this study are available from the corresponding author upon reasonable request.

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